GigaWhat Energy

GigaWhat Gas
Controlled procedure

PR-GAS-040 Working in Hazardous Areas (DSEAR Zones)

Field	Value
Document ID	PR-GAS-040
Version	3
Effective from	2024-06-03
Review due	2027-06-03
Owner	Gas Safety Manager
Approved by	Head of Gas Network Operations
Applies to	All GigaWhat Gas sites and all gas asset classes

FICTIONAL DOCUMENT. Created for the GigaWhat demo. Not for real operational use.

1 Purpose

This procedure sets out how GigaWhat Gas staff and contractors work safely in places where a flammable gas atmosphere may be present. It explains the hazardous area zones used at GigaWhat Energy sites, the equipment allowed in each zone, and the controls for static, hot work and gas detector alarms.

2 Scope

This procedure applies to all work in permanent hazardous areas at GPR-GWT, GPR-MLN, GPR-STN and GPR-KAV, and in temporary hazardous areas created by venting, purging, open pipework or gas escapes anywhere on the GigaWhat gas network.

3 References

- PR-GAS-003 Response to Reported Gas Escapes
- PR-GAS-021 Permit-to-Work for Gas Operations
- PR-GAS-031 Purging and Commissioning of Polyethylene Mains
- PR-CORP-001 Reporting Incidents and Near Misses
- Dangerous Substances and Explosive Atmospheres Regulations 2002 (DSEAR) and HSE guidance L138
- Gas Safety (Management) Regulations 1996 and HSE guidance L80
- Management of Health and Safety at Work Regulations 1999

- HSE guidance HSG250 (permit-to-work systems)

4 Definitions

Hazardous area: A place where a flammable gas atmosphere may be present in enough quantity to need special precautions.

Zone 0: An area where a flammable atmosphere is present continuously or for long periods. At GigaWhat Energy this is limited to the inside of pipework and vessels.

Zone 1: An area where a flammable atmosphere is likely to occur occasionally in normal operation.

Zone 2: An area where a flammable atmosphere is not likely in normal operation and, if it does occur, will last only a short time.

Ex equipment: Equipment certified for use in a hazardous area, marked with its equipment category, gas group and temperature class.

Gas-free certificate: A record, issued by an Authorised Person (Gas), that gas tests show no flammable gas at a defined location for a defined time.

5 Responsibilities

Gas Safety Manager: Owns the DSEAR risk assessments and hazardous area classification drawings for each site, and reviews them every 3 years or after any change to the plant.

Authorised Person (Gas): Identifies hazardous areas when planning work, issues gas-free and hot work certificates under PR-GAS-021, and releases each hold point.

Competent Person (Pressure Systems): Follows the zone controls, carries out gas tests and stops work when alarms sound.

First Call Operative: Applies the temporary hazardous area controls in 6.5 at reported escapes.

Gas Network Controller: Is told when anyone enters or leaves a permanent hazardous area at an unmanned site.

6 Procedure

6.1 Hazardous areas at GigaWhat Energy sites

The hazardous area classification drawing for each site is kept in the site document box and in the asset file. The table summarises the zones.

Site	Zone 1	Zone 2
GPR-GWT	Within 1 m of the creep relief, slam-shut and odorant tank vent terminals	Within 3 m of any flange, valve or the filter FLT-GWT-01 inside the compound
GPR-MLN, GPR-STN, GPR-KAV	Within 1 m of governor, creep relief and slam-shut vent terminals	The whole inside of the governor kiosk, and 1 m around open kiosk doors

> NOTE: Use the site drawing, not this summary, when planning work. If the drawing and the plant do not match, stop and tell the Authorised Person (Gas).

6.2 Before entering a zone

6.2.1 Read the hazardous area classification drawing and identify the zones at the work location.

6.2.2 Wear flame-retardant, antistatic clothing and safety footwear.

6.2.3 Switch on your personal gas detector in clean air, confirm it has passed its bump test, and wear it in the breathing zone.

6.2.4 Switch off any non-Ex mobile phone, radio or electronic device and leave it outside the zone.

6.2.5 Park vehicles outside the zone and switch off engines within 5 m of any zone boundary.

6.2.6 At unmanned sites, tell the Gas Network Controller when you enter and leave.

6.3 Equipment and tools

> WARNING: Non-certified electrical equipment can ignite a gas atmosphere. Bringing it into a zone without a gas-free certificate can cause an explosion.

6.3.1 Use only Ex equipment suitable for the zone: equipment category 2 or better in Zone 1, and category 3 or better in Zone 2, rated for gas group IIA and temperature class T1 or better.

6.3.2 Check the Ex marking and the inspection label before use, and do not use damaged equipment.

6.3.3 Use hand tools only. Do not use power tools, grinders or cutting equipment unless they are Ex certified or covered by a hot work certificate.

6.3.4 Do not bring non-Ex equipment into a zone unless the Authorised Person (Gas) has issued a gas-free certificate for that location and period.

> HOLD POINT: Non-Ex equipment must not enter a zone until the Authorised Person (Gas) has confirmed gas-free conditions and issued the certificate.

6.4 Static and stray electrical currents

> WARNING: Separating metallic pipework can draw a spark from cathodic protection or stray currents. Gas flowing through polyethylene pipe can build up a static charge that discharges as a spark.

6.4.1 Before cutting or separating any metallic pipe or fitting, fit a temporary continuity bond across the break and leave it in place until the joint is remade.

6.4.2 On polyethylene pipe, wet the pipe surface and wrap it with a wet cloth connected to a driven earth spike before cutting or venting.

6.4.3 Earth all metallic vent stacks with a driven earth spike.

6.4.4 Do not wipe polyethylene pipe with a dry cloth.

6.5 Temporary hazardous areas

6.5.1 Treat the area within 3 m of any open pipework, venting point or gas escape as Zone 1 until gas tests show no gas.

6.5.2 For purging, apply the vent stack and exclusion zone requirements of PR-GAS-031.

6.5.3 At reported escapes, treat any building or excavation where gas has been detected as Zone 1 and follow PR-GAS-003.

6.5.4 Mark temporary zones with barriers and signs and keep the public out.

6.5.5 Re-test temporary zones at least every 15 minutes while work continues.

6.6 Hot work in or near a zone

> WARNING: Hot work in a gas atmosphere can cause fire or explosion. Never start hot work on the strength of a gas test taken earlier in the day.

6.6.1 Obtain a permit-to-work and hot work certificate from the Authorised Person (Gas) under PR-GAS-021.

6.6.2 Positively isolate the plant and prove it gas-free under PR-GAS-021.

6.6.3 Test the work location and the surrounding 3 m with a calibrated gas detector immediately before work starts, and confirm no gas is detected.

6.6.4 Keep a gas detector running at the work location throughout.

6.6.5 Keep a dry powder extinguisher at the work and a fire watch in place for 30 minutes after hot work stops.

> HOLD POINT: Do not start hot work until the Authorised Person (Gas) has witnessed the gas test, signed the hot work certificate and confirmed the fire precautions are in place.

6.7 Gas detector alarms

6.7.1 First alarm, 10% LEL: stop any hot work, stop work that may release gas, ventilate and find the source.

6.7.2 Second alarm, 20% LEL: leave the zone immediately by the shortest safe route and do not re-enter.

6.7.3 After a second alarm, make the plant safe from outside the zone if possible and tell the Authorised Person (Gas) and the Gas Network Controller.

6.7.4 Re-enter only when the Authorised Person (Gas) has confirmed readings are below 10% LEL and the source is controlled.

6.8 Emergency

6.8.1 Call 999 for fire, explosion or injury.

6.8.2 Report uncontrolled gas escapes to the Gas Network Controller, or to the National Gas Emergency Service on 0800 111 999 if the Controller cannot be reached.

6.8.3 Report every second alarm, ignition and near miss under PR-CORP-001.

7 Records

Hazardous area classification drawings and DSEAR risk assessments are kept in the asset file and the site document box, with their review dates. Gas-free and hot work certificates are kept with the permit-to-work for 6 years. Ex equipment inspection records are kept for the life of the equipment.

8 Revision history

Version	Date	Change
1	2015-06-01	First issue.
2	2020-06-01	Added static and stray current controls and temporary hazardous areas.
3	2024-06-03	Added zone summary table for each site, including the GPR-GWT odorant tank vent. Added gas detector alarm actions and hold point before non-Ex equipment enters a zone.